

13.1 The multiple testing problem

Seven groups give 21 pairs, and a two-sided t test at level 0.05 for each pair is a correct test. But an analyst who runs all 21 and reports the significant ones is not using a procedure with error rate 0.05: even if all seven means are equal, about one pair is expected to be significant, and the reader cannot tell it from a real difference. This section defines the error rates of a family of inferences, measures by simulation how large they become without adjustment, and separates strong control from a weak form that protects only against the situation in which nothing is going on.

13.1.1 A family of hypotheses

Let $H_1, \dots, H_m$ be null hypotheses about the parameters of a model, tested on the same data. The model's true parameter makes some of them true and the others false. Write $I_0 \subseteq \{1, \dots, m\}$ for the set of true hypotheses and $m_0 = |I_0|$ for their number. A **multiple testing procedure** maps the data to a set $\mathcal{R} \subseteq \{1, \dots, m\}$ of rejected hypotheses. The outcome can be summarized by three counts:

	not rejected	rejected	total
true hypotheses	$m_0 - V$	V	m_0
false hypotheses	$m - m_0 - S$	S	$m - m_0$
total	$m - R$	R	m

Here $R = |\mathcal{R}|$ is the number of rejections, $V = |\mathcal{R} \cap I_0|$ the number of **false rejections** (Type I errors), and $S = R - V$ the number of correct ones. The counts R , V and S are random, and m_0 is unknown. The analyst sees R but not how it splits into V and S .

With one true hypothesis, a level- α test makes $\Pr(V = 1) \leq \alpha$. For a family there are several reasonable ways to summarize the distribution of V , and they lead to different procedures.

Definition 13.1.1. Error rates of a family

For a multiple testing procedure and a given value of the parameters:

- (a) the **familywise error rate** is $\text{FWER} = \Pr(V \geq 1)$, the probability of at least one false rejection;
- (b) the **per-family error rate** is $\text{PFER} = \mathbb{E}(V)$, the expected number of false rejections, and the **per-comparison error rate** is $\text{PCER} = \mathbb{E}(V)/m$;
- (c) the **false discovery proportion** is $Q = V/\max(R, 1)$, which is V/R when something is rejected and 0 otherwise, and the **false discovery rate** is its expectation, $\text{FDR} = \mathbb{E}(Q)$.

A procedure **controls** one of these rates at level α if the rate is at most α for every value of the parameters.

The familywise error rate treats one false claim as a failure of the whole family; it suits conclusions that will each be acted on. The false discovery rate accepts some errors in a long list of discoveries and bounds their expected share. The unadjusted procedure, testing each hypothesis at level α , controls only the per-comparison rate.

Proposition 13.1.1. How the error rates compare

For every procedure and every value of the parameters,

$$\text{PCER} \leq \text{FWER} \leq \text{PFER}, \quad \text{FDR} \leq \text{FWER}.$$

If all the hypotheses are true ($m_0 = m$), then $\text{FDR} = \text{FWER}$. Consequently a procedure that controls the per-family error rate at α controls the familywise rate at α , and a procedure that controls the familywise rate controls the false discovery rate.

Proof. All four rates are expectations of functions of V and R , so it is enough to compare the functions pointwise. Since $0 \leq V \leq m$, we have $V/m \leq 1_{\{V \geq 1\}} \leq V$, which gives the first chain after taking expectations. Since $V \leq R$, the ratio $V/\max(R, 1)$ is at most 1, and it is 0 when $V = 0$; so $V/\max(R, 1) \leq 1_{\{V \geq 1\}}$, which gives $\text{FDR} \leq \text{FWER}$. If every hypothesis is true, every rejection is false, so $V = R$ and $V/\max(R, 1) = 1_{\{V \geq 1\}}$ exactly. ■

In particular, a procedure that controls the false discovery rate is a valid level- α test of the hypothesis that *all* the H_j are true, but not in general a procedure with familywise control (Section 13.5).

13.1.2 How often something is significant by chance

If the m tests are independent and each has exact level α , then when all hypotheses are true,

$$\text{FWER} = 1 - \Pr(\text{no rejection}) = 1 - (1 - \alpha)^m. \quad (13.1.1)$$

With $\alpha = 0.05$ this is 0.401 for ten tests and 0.994 for a hundred. In a linear model the tests are rarely independent. They share the estimate s of σ , and comparisons involving a common group share that group's mean. The next example measures the effect of that dependence for the most common family of all.

Example. All pairs of equal means

Take g groups of 5 observations each, with all group means equal, and compare every pair with an unadjusted two-sided t test at level 0.05, using the pooled variance estimate on $g(5 - 1)$ degrees of freedom. The listing simulates 20000 data sets for each g . For $g = 3, 5, 7, 10$ (so 3, 10, 21, 45 pairs), the proportion of data sets with at least one significant pair is

$$0.116, \quad 0.265, \quad 0.412, \quad 0.589.$$

Independent tests would give 0.143, 0.401, 0.659 and 0.901. The average number of significant pairs is 0.15, 0.50, 1.08 and 2.23, in each case 0.05 times the number of pairs up to simulation error. Figure 13.1.1 plots the familywise rate against the number of tests.

```
import itertools
import numpy as np
from scipy import stats
```

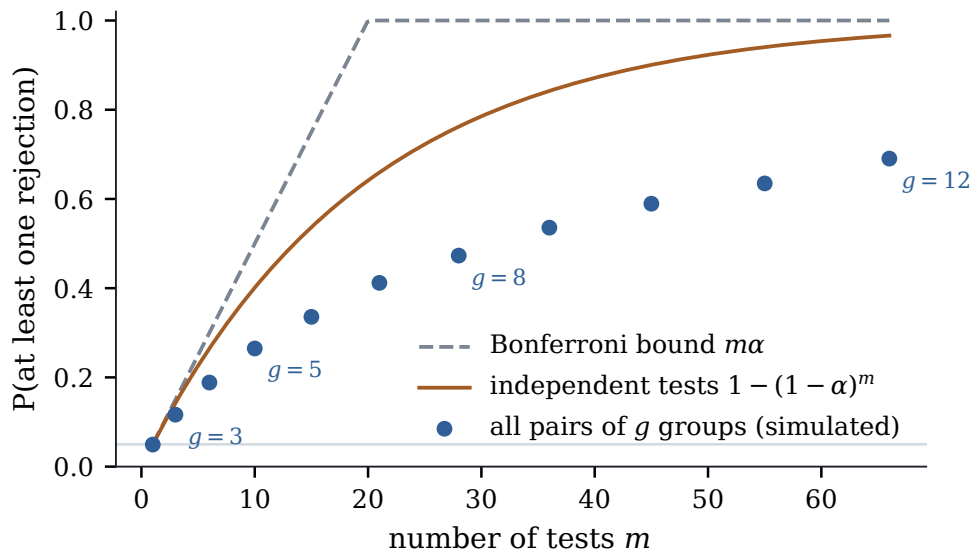


Figure 13.1.1: Probability of at least one false rejection among m unadjusted level-0.05 tests: all pairs of $g = 2, \dots, 12$ equal means (dots, simulated), independent tests (curve, Eq. 13.1.1) and the Bonferroni bound $m\alpha$ (dashed).

```
alpha, reps, m_rep = 0.05, 20_000, 5                                # 5 observations per group

def any_significant(g, m_rep, reps, alpha):
    """Share of data sets with at least one unadjusted pairwise rejection, and mean
    count."""
    rng = np.random.default_rng([20260919, g])                      # one fixed stream per g
    nu = g * (m_rep - 1)                                             # error degrees of freedom
    y = rng.normal(size=(reps, g, m_rep))                           # all group means equal
    means = y.mean(axis=2)
    s2 = ((y - means[:, :, None]) ** 2).sum(axis=(1, 2)) / nu
    se = np.sqrt(s2 * 2 / m_rep)                                     # standard error of a
    difference
    crit = stats.t.ppf(1 - alpha / 2, nu)
    count = np.zeros(reps)
    for k, l in itertools.combinations(range(g), 2):
        count += np.abs(means[:, k] - means[:, l]) / se > crit
    return np.mean(count > 0), np.mean(count)

for g in (3, 5, 10):
    pairs = g * (g - 1) // 2
    fwer, mean_count = any_significant(g, m_rep, reps, alpha)
    print(f"g = {g:2d}, {pairs:2d} pairs: P(some rejection) = {fwer:.3f}, "
          f"mean number = {mean_count:.2f} (alpha x pairs = {alpha * pairs:.2f}), "
          f"independent tests would give {1 - (1 - alpha) ** pairs:.3f}")
```

[Run this code online](#)

The expected number of false rejections is additive whatever the dependence, which is why the Bonferroni bound of Section 13.4 needs no assumptions. The probability of at least one grows more slowly than for independent tests, because false rejections cluster when s is small or one mean is extreme. Still, with ten groups some pair is “significant” in well over half of all experiments in which nothing differs.

13.1.3 Simultaneous intervals and the tests they induce

Let ψ_j , $j \in J$, be real parameters, where the index set J may be infinite. Intervals $[L_j, U_j]$ computed from the data are **simultaneous confidence intervals** at level $1 - \alpha$ if

$$\Pr(L_j \leq \psi_j \leq U_j \text{ for every } j \in J) \geq 1 - \alpha$$

for every value of the parameters. Scheffé's and Tukey's methods and the Bonferroni intervals all produce them, and they yield tests with the strongest guarantees.

Proposition 13.1.2. Tests from simultaneous intervals

Let $[L_j, U_j]$, $j \in J$, be simultaneous confidence intervals at level $1 - \alpha$, and for given constants d_j reject $H_j : \psi_j = d_j$ exactly when $d_j \notin [L_j, U_j]$. Then:

- (a) the probability of rejecting at least one true H_j is at most α , for every value of the parameters;
- (b) if, whenever H_j is rejected, one also declares $\psi_j > d_j$ or $\psi_j < d_j$ according to the side of d_j on which the interval lies, the probability of at least one false rejection *or* wrong declaration of sign is at most α .

Proof. (a) If a true H_j is rejected, then $\psi_j = d_j \notin [L_j, U_j]$, so some interval fails to cover. That event has probability at most α . (b) A wrong declaration, say $\psi_j > d_j$ when in fact $\psi_j \leq d_j$, happens only if $L_j > d_j \geq \psi_j$; again an interval fails to cover. Both kinds of error are contained in the event that some interval misses, whose probability is at most α . ■

Part (b) matters because few parameters are exactly equal to their hypothesized values; the real question is which way each difference goes. Some good procedures, such as Holm's, give tests but no simple companion intervals.

13.1.4 Weak control is not enough

The error rates depend on which hypotheses are true. The strongest requirement bounds the rate in every configuration; a much weaker one only when every hypothesis is true.

Definition 13.1.2. Strong and weak control

A procedure controls the familywise error rate **strongly** at level α if $\Pr(V \geq 1) \leq \alpha$ for every value of the parameters, and **weakly** if $\Pr(V \geq 1) \leq \alpha$ whenever all of $H_1, \dots, H_m$ are true.

Weak control is easy to obtain. Let $H_0 = \bigcap_j H_j$ be the **global null hypothesis** that every H_j holds.

Proposition 13.1.3. A gate gives weak control

Let ϕ_0 be a level- α test of the global null H_0 . A procedure that rejects no H_j unless ϕ_0

rejects H_0 controls the familywise error rate weakly at level α .

Proof. When all H_j are true, $V \geq 1$ requires a rejection by ϕ_0 , which has probability at most α because H_0 is true. ■

The best-known gated procedure is Fisher's **protected least significant difference** (LSD): in a one-way layout, test equality of all g means with the F test at level α , and only if it rejects, compare each pair with an unadjusted t test at level α . Weak control is all it has.

Example. The protected LSD without strong control

With $g = 10$ groups of 5 observations and all means equal, the simulation below gives a familywise error rate of 0.049, as [Proposition 13.1.3](#) promises. Now move one mean far from the other nine, which remain equal. The F test rejects in essentially every data set, the gate is always open, and the 36 pairwise comparisons among the nine equal means are made without adjustment. The familywise error rate rises to 0.539.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps, m_rep = 0.05, 20_000, 5

def protected_lsd_fwer(mu, m_rep, reps, alpha, seed):
    """Share of data sets in which the protected LSD rejects some true equality."""
    rng = np.random.default_rng(seed)
    g = len(mu)
    nu = g * (m_rep - 1)
    y = np.asarray(mu)[None, :, None] + rng.normal(size=(reps, g, m_rep))
    means = y.mean(axis=2)
    s2 = ((y - means[:, :, None]) ** 2).sum(axis=(1, 2)) / nu
    between = m_rep * ((means - means.mean(axis=1, keepdims=True)) ** 2).sum(axis=1)
    gate = between / (g - 1) / s2 > stats.f.ppf(1 - alpha, g - 1, nu)  # the F test
    crit = stats.t.ppf(1 - alpha / 2, nu) * np.sqrt(2 * s2 / m_rep)
    false = np.zeros(reps, dtype=bool)
    for k, l in itertools.combinations(range(g), 2):
        if mu[k] == mu[l]:  # a true null hypothesis
            false |= gate & (np.abs(means[:, k] - means[:, l]) > crit)
    return false.mean()

all_equal = protected_lsd_fwer([0.0] * 10, m_rep, reps, alpha, seed=1)
one_apart = protected_lsd_fwer([0.0] * 9 + [10.0], m_rep, reps, alpha, seed=2)
print(f"10 groups, all means equal:      FWER = {all_equal:.3f}")
print(f"10 groups, one mean far from rest: FWER = {one_apart:.3f}")
```

[Run this code online](#)

Weak control protects only against the one configuration in which there is nothing to find. Every procedure recommended in this chapter controls its error rate strongly.

13.1.5 Exercises

13.1.5.1 A. Check your understanding

Exercise. A1

Ten and twenty independent tests of true hypotheses are made, each at level 0.01. Find the familywise error rate in each case. What common level for twenty independent tests gives a familywise rate of exactly 0.05?

Solution. By [Eq. 13.1.1](#) the rates are $1 - 0.99^{10} = 0.0956$ and $1 - 0.99^{20} = 0.1821$. The level α with $1 - (1 - \alpha)^{20} = 0.05$ is $\alpha = 1 - 0.95^{1/20} = 0.00256$, slightly more than $0.05/20 = 0.00250$. This is the Šidák level of [Section 13.4](#).

Exercise. A2

Give a configuration of true and false hypotheses, and a procedure, for which the familywise error rate is close to one while the false discovery rate is close to zero.

13.1.5.2 B. Practice

Exercise. B1

Show that with $g = 3$ groups the protected LSD controls the familywise error rate strongly. *Hint:* consider separately the configurations with three, one and no true equalities among the three pairwise hypotheses.

Solution. The three hypotheses are $\mu_1 = \mu_2$, $\mu_1 = \mu_3$, $\mu_2 = \mu_3$. If two of them are true, the third is true as well, so the possible numbers of true hypotheses are 3, 1 and 0. With three, all means are equal and weak control ([Proposition 13.1.3](#)) applies. With none, no false rejection is possible. With exactly one, say $\mu_1 = \mu_2 \neq \mu_3$, a false rejection needs the unadjusted t test of $\mu_1 = \mu_2$ to reject, an event of probability α whether or not the gate opens. So the familywise rate is at most α in every configuration. With four or more groups, a configuration can hold two disjoint true equalities, and the argument fails.

Exercise. B2

Let m_0 true hypotheses have independent uniform p -values, and let m_1 false hypotheses have p -values equal to 0. The unadjusted procedure rejects when $p_j \leq \alpha$. Find its familywise error rate, show that its false discovery rate is at most $m_0\alpha/m_1$, and conclude that the false discovery rate tends to 0 as $m_1 \rightarrow \infty$ with m_0 fixed.

13.1.5.3 C. Going deeper

Exercise. C1

In a one-way layout with $g \geq 3$ groups of m observations and known σ , run the protected

LSD with z tests: the gate is the level- α test that rejects when $\sum_k m(\bar{y}_k - \bar{y})^2/\sigma^2$ exceeds the upper α point of $\chi^2(g-1)$, and a pair is declared different when $|\bar{y}_k - \bar{y}_l| > z_{\alpha/2} \sigma \sqrt{2/m}$, where $z_{\alpha/2}$ is the upper $\alpha/2$ point of $N(0, 1)$. Let $\mu_1 = \dots = \mu_{g-1} = 0$ and $\mu_g = \Delta$. Show that as $\Delta \rightarrow \infty$ the familywise error rate tends to $\Pr(W_{g-1} > \sqrt{2} z_{\alpha/2})$, where W_{g-1} is the range of $g-1$ independent $N(0, 1)$ variables. Show that this limit is at least $1 - (1 - \alpha)^{\lfloor (g-1)/2 \rfloor}$ and that it tends to 1 as $g \rightarrow \infty$.

Solution. With $Z_k = \sqrt{m} \bar{y}_k/\sigma$, $k < g$, independent $N(0, 1)$ and free of Δ , some pair $k, l < g$ is declared different iff $W_{g-1} = \max_{k < g} Z_k - \min_{k < g} Z_k > \sqrt{2} z_{\alpha/2}$, so the rate is $\Pr(G \cap \{W_{g-1} > \sqrt{2} z_{\alpha/2}\})$, with G the event that the gate opens. The gate statistic is at least $m(\bar{y}_g - \bar{y})^2/\sigma^2$, where $\bar{y}_g - \bar{y}$ is $\frac{g-1}{g}\Delta$ plus a variable free of Δ ; so $\Pr(G) \rightarrow 1$, giving the limit. The disjoint pairs $(1, 2), (3, 4), \dots$ give $\lfloor (g-1)/2 \rfloor$ independent statistics $|Z_{2i-1} - Z_{2i}|/\sqrt{2}$, each exceeding $z_{\alpha/2}$ with probability α , and W_{g-1} is at least $\sqrt{2}$ times each of them; hence the lower bound, which tends to 1.